APPENDIX C

Time

This chapter introduces time scales that are used in astronomical and spacecraft work. Time is needed to determine the orientation of a planet with respect to the inertial frame. It is also needed for star cameras to accurately determine the azimuth and elevation of stars in the star field. For deep-space spacecraft, time is needed to determine the positions of the planets relative to the spacecraft.

C.1. Time scales

A time scale is defined by a specific periodic natural phenomenon. The formal definition must include a description of the phenomenon to be used (i.e., what defines a period), the rate of advance (the ratio of time units to the natural period) and an initial epoch, which is a time reading of some identifiable event.

There are two time scales used in astronomy. The first is based on the "atomic" second in Systeme International (SI) with an SI second corresponding to 9 192 631 770 cycles of the radiation corresponding to the ground-state hyperfine transition of Cesium 133. The second is based on the rotation of the Earth. Since the rotation of the Earth is not constant, the relationship between the two time scales is complex.

Important time scales are:

- UT1 (Universal Time 1)—A time scale based on the rotation of the Earth. UT1 is defined with respect to Sidereal Time.
- TT (Terrestrial Time)—The theoretically ideal time scale at the surface of the Earth.
- TDT (Terrestrial Dynamical Time)—A time scale that would be kept by an ideal clock at sea level.
- TDB (Barycentric Dynamical Time)—A time scale that would be kept by an ideal clock moving with the solar-system barycenter. It is always within 2 milliseconds of TDT and the difference is caused by relativistic effects.
- TAI (International Atomic Time)—A time scale kept by atomic clocks on the Earth.
- UTC (Universal Time Coordinated)—The basis for international time keeping. Its rate is the same as TAI, but its epoch is adjusted in one-second steps to keep it within 0.9 s of UT1, which is measured by the Earth's rotation against the stars.
- DUT1 (Universal Time Correction)—The difference between UT1 and UTC.
- Sidereal Time—Measured by the rotation of the Earth against the stars.

The standard epoch for modern astrometric reference data, designated J2000.0, is expressed as a Terrestrial Time TT instant: J2000.0 is 2000 January 1, 12h TT (JD 2451545.0 TT) at the geocenter.

Universal time (UT) now always refers to UT1. UT1 is a linear function of the Earth's rotation angle θ, which is the geocentric angle between two directions in the equatorial plane called the Celestial Intermediate Origin (CIO) and the Terrestrial Intermediate Origin (TIO). The TIO rotates with the Earth but CIO does not, so that $\dot{\theta} = \omega$, the Earth's angular rate. However, ω is not a constant and must be measured through astronomical observations.

The worldwide system of civil time is based on Coordinated Universal Time (UTC). UTC uses the SI second on the geoid as its fundamental unit, but is subject to occasional 1-second adjustments to keep it within 0.9 second of UT1. Such adjustments, called leap seconds, are normally introduced at the end of June or December, when necessary, by international agreement.

$$TAI = UTC + \Delta AT \tag{C.1}$$

where ΔAT is the accumulated number of leap seconds. The astronomical time scale Terrestrial Time is

$$TT = TAI + 32.184s \tag{C.2}$$

Geocentric coordinate time is

$$TCG = \frac{TT - L_G t_0}{1 - L_G} \tag{C.3}$$

where $L_G = 6.969290134 \times 10^{-10}$.

TT, TCG, and TCB all read 1977 January 1 00:00:32.184 (JD 2443144.5003725) on 1977 January 1, 00:00:00 TAI (JD 2443144.5 TAI) at the geocenter. This event is t_0 in the above equations.

The relationship between TDB and TT is more complex but can be approximated by the series

$$\begin{aligned} TDB \approx TT &+ 0.001657\sin(628.3076T + 6.2401) \\ &+ 0.000022\sin(575.3385T + 4.2970) \\ &+ 0.000014\sin(1256.6152T + 6.1969) \\ &+ 0.000005\sin(606.9777T + 4.0212) \\ &+ 0.000005\sin(52.9691T + 0.4444) \\ &+ 0.000002\sin(21.3299T + 5.5431) \\ &+ 0.000010\sin(628.3076T + 4.2490) + \ldots \end{aligned} \tag{C.4}$$

where T is the number of Julian centuries from J2000.0 or

$$T = \frac{JD(TT) - 2451545.0}{36525} \tag{C.5}$$

The maximum error using the above expansion is 10 μs between the years 1600 and 2200, which is two orders of magnitude better than $TT = TDB$.

$$UT1 = UTC + (UT1 - UTC) \tag{C.6}$$

or

$$UT1 = UTC + DUT1 \tag{C.7}$$

where DUT1 is a broadcast approximation to UT1-UTC, which is accurate to 0.1 second,

$$UT1 = UTC - \Delta T \tag{C.8}$$

where

$$\Delta T 32.184 + \Delta AT - (UT1 - UTC) \tag{C.9}$$

Values of ΔT are listed in the Astronomical Almanac [1].

C.2. Earth rotation

The Earth rotates about its axis. This rotation is measured with respect to the inertial frame. The terrestrial day is referenced to Earth rotation.

The Earth's rotation angle, θ, is

$$\theta = 0.7790572732640 + 1.00273781191135448 D_U \tag{C.10}$$

where

$$D_U = JD(UT1) - 2451545.0. \tag{C.11}$$

A more accurate formula is

$$\theta = 0.7790572732640 + 0.00273781191135448 D_U + \mathrm{frac}(JD(UT1)) \tag{C.12}$$

Greenwich Mean Time (GMT) is

$$\begin{aligned} GMST = 86400\ \theta + (0.014506 + 4612.156534\ T + 1.3915817\ T^2 - 0.00000044\ T^3 \\ - 0.000029956\ T^4 - 0.0000000368\ T^5)/15 \end{aligned} \tag{C.13}$$

where T is the number of centuries in TDB (or TT). Sidereal time is the time used by astronomers to locate celestial objects. The apparent sidereal time is

$$GAST = GMST + \frac{\epsilon_T}{15} \tag{C.14}$$

where ϵ_T is the Equation of the Equinoxes, which is approximately

$$
\begin{aligned}
E = {} & \psi \cos\epsilon && \text{(C.15)}\\
& + 0.00264096 \sin(\Omega) && \text{(C.16)}\\
& + 0.00006352 \sin(2\Omega)\\
& + 0.00001175 \sin(2F - 2D + 3\Omega)\\
& + 0.00001121 \sin(2F - 2D + \Omega)\\
& - 0.00000455 \sin(2F - 2D + 2\Omega)\\
& + 0.00000202 \sin(2F + 3\Omega)\\
& + 0.00000198 \sin(2F + \Omega)\\
& - 0.00000172 \sin(3\Omega)\\
& - 0.00000087 T \sin(\Omega) + \ldots
\end{aligned}
$$

where ψ is the nutation in longitude, in arcseconds; ϵ is the mean obliquity of the ecliptic; and F and D, are fundamental lunisolar arguments. Mean sidereal time does not use the Equation of the Equinoxes.

Local sidereal time is

$$LXST = GXST + \frac{3600}{15}\lambda \tag{C.17}$$

where X is A or M and λ is the longitude of the place of interest in degrees.

C.3. Julian date

Julian date gives a continuous count of dates since January 1, 4713 BC at noon. Integer Julian Dates always refer to noon. The Julian date for the standard epoch J2000.0, January 1, 2000 12h is JD 2451545.0 TDB.

Julian date is very convenient in control work because it does not require any complex logic, as would years and days. However, to get high accuracy you need a lot of digits. Accuracy to the day requires 7 digits. Accuracy to the millisecond requires another 9 digits. Sometimes Modified Julian Date (MJD) is used instead. The MJD begins at midnight rather than noon.

$$MJD = JD - 2400000.5 \tag{C.18}$$

With double-precision numerics on modern-day processors, MJD is not very important.

Knowledge of the Julian date is also required when transforming between Earth-Centered Inertial (ECI) and Earth-Fixed (EF) coordinate frames. Earth-Fixed coordinate frames rotate with the Earth, and the Julian date is required in order to compute the amount of rotation that has occurred since the last reference epoch.

Another important time is the Greenwich Mean Sidereal Time, which is the angle between the ECI x-axis (Vernal Equinox) and the Greenwich Meridian.

C.4. Time standards

The following sections discuss terrestrial time standards. If you have a vehicle on another planet then your day will be determined by the rotation of that planet.

C.4.1 Local time

The date/time that you see on local clocks or on the Internet.

C.4.2 UTC

Coordinated Universal Time, more precisely known as GMT (Greenwich Mean Time), or Zulu time. Local times around the world differ by time zones. For example, Eastern Standard Time is exactly one hour later than Atlantic time. Some time zones have standard and daylight times. There are currently 37 different time zones in use around the world.

C.4.3 GPS

Global Positioning System time, is the time scale implemented by the atomic clocks in the GPS ground-control stations and the GPS satellites. GPS time was zero at midnight January 6, 1980.

Table C.1 Planetary days.

Celestial Object	Day (Earth Hours)
Mercury	1408
Venus	5832
Earth	24
Mars	25
Jupiter	10
Saturn	11
Uranus	17
Neptune	16
Pluto	153
Europa	84 (tidally locked to its orbit period)
Enceladus	32.9 (tidally locked to its orbit period
Titan	382 (tidally locked to its orbit period)
Moon	672 (tidally locked to its orbit period)

C.4.4 Loran-C

Long-Range Navigation time, is a time scale measured by the atomic clocks in Loran-C chain transmitter sites. Loran time was zero at midnight January 1, 1958.

C.4.5 TAI

Temps Atomique International, is the international atomic time scale based on a continuous counting of the International System of Units second. TAI leads GPS time by 19 seconds.

C.4.6 Planetary days

Table C.1 lists planetary day lengths for selected solar-system objects.

References

[1] P.K. Seidelmann, Explanatory Supplement to the Astronomical Almanac, University Science Books, Mill Valley, CA, 1992.